

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	C++, Python, JavaScript, TypeScript, SQL, Bash
Systems & Concurrency	Multithreading, async services / queues, OpenMP / SIMD concepts, Linux profiling, performance optimization, Spark / PySpark, Spark UI
Backend & APIs	API design, REST APIs, FastAPI, Node.js, clean contracts / abstraction boundaries, multi-tenant SaaS, production debugging
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux
Databases & Tools	PostgreSQL, MongoDB, Redis, SQLite, GTest / CMake

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability and security for **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Worked cross-functionally with clinical and product teams to monitor production logs, investigate incident bugs, and ship fixes for tenant-specific and integration issues during releases; improved operational stability through authentication hardening and rate limiting.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify over **PostgreSQL** with clear contracts, input validation, appropriate **abstraction boundaries**, and backward-compatible changes, building simple interfaces over backend complexity used in production by the frontend team.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python

- Cut end-to-end Spark ETL runtime roughly **25%** (17 to 19 min down to 14.3 min) on a **1.4B-row** dataset by replacing shuffle joins with broadcast joins, enabling adaptive query execution and skew-join handling, and tuning shuffle partitions and partitioning.
- Evaluated performance via **Spark UI** stage-level profiling to localize shuffle and skew bottlenecks; applied column pruning and precomputed fields to cut data movement, and documented a repeatable optimization workflow.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Engineered FastAPI backend and Electron/React frontend running **fully on-device**; designed async APIs and queues to handle **concurrent** video and audio streams while keeping the UI responsive, with no external network dependencies.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and tool failures; conversation state persists server-side in D1 without an external database.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, and verifies outputs, decomposing complex behavior into clear, testable stages with explicit boundaries.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation for review-friendly debuggability.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.